

Project Design Phase-II

Data Flow Diagram & User Stories

Date	2 November 2025
Team ID	NM2025TMID05911
Project Name	Laptop Request Catalog Item
Maximum Marks	4 Marks

Data Flow Diagram:

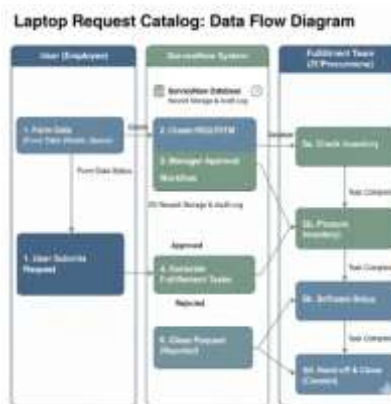
- A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.
- In the project "Laptop Request Catalog", the Data Flow Diagram (DFD) illustrates how the laptop request data is processed, validated, routed for approval, and ultimately used to drive the fulfillment process. The DFD ensures that the data collected from the user is transformed and stored correctly in the ServiceNow Database to maintain accountability and drive subsequent automated tasks.
- The DFD shows the interaction between the User, the central ServiceNow System (acting as the processor), and the Fulfillment Team (the external entity executing the final steps).

Example:



1. **Employee Submits Request:** The user fills out the dynamic Service Catalog form with their laptop specifications (model, RAM, storage) and clicks submit.

2. **Manager Approval:** The system automatically sends the request to the manager. The process **pauses** until the manager approves or rejects the request.
3. **Fulfillment Task 1: Check Inventory:** If approved, the first task is created and assigned to IT Fulfillment to check the physical stock of the requested laptop model.
4. **Fulfillment Tasks 2 & 3 (Sequential):** Based on the inventory check, the flow either creates a **Procurement** task (if out of stock) or skips directly to the **Setup & Tagging** task (to install software and prep the device).
5. **Request Closure:** Once the final setup task is complete, the system sends an automated "Ready for Pickup" notification to the employee and sets the final request record status to Closed Complete.



User Stories:

User stories define what different users need from the system in simple, goal-focused language, ensuring the system meets specific acceptance criteria.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Requester (Employee)	Catalog Usability & Efficiency	US-E1	As an employee , I want the request form fields (e.g., RAM, Storage) to dynamically appear or change based on the selected Laptop Model so I only see relevant options.	Dynamic fields must automatically populate or hide based on the Laptop Model selection using Client Scripts and/or UI Policies.	High	Sprint-1

Approver (Manager)	Approval Process	US-A1	As a Manager , I want to receive a clear email notification when a direct report submits a request, including all the laptop specifications, so I can approve or reject quickly.	The notification must contain a deep link to the Approval record and a summary of all RITM variables.	High	Sprint-1
Fulfillment Team (IT)	Task Automation & Contro	US-F1	As an IT Fulfillment Technician , I want the workflow to automatically generate sequential tasks (Inventory $\rightarrow$ Setup) so I complete steps in the required order.	The next task must not be assigned until the preceding Catalog Task is marked 'Closed Complete	High	Sprint-2
Governance	Audit and Data Integrity	US-G3	As an Auditor , I want all approval history, task completion times, and variable changes to be automatically recorded and viewable on the RITM record, to maintain governance.	The Activity Stream on the RITM must log all state changes, approval decisions, and task transitions.		